

ASM CVD Tubechange		User ID:	Luke
Date: 26.08.2026	PM Art: ASM_Tubechange	System ID:	QD542
Start Time: 8:00:00 am		End Time:	
Preparation of the Tube Change	Reference Data	Data	Comments
Prepare all parts for tube change. Inform Operators that your taking the tube for PM Do a Leak Check for Reference Put reactor into Maintenance mode and download maint recipe. Start Maint recipe, use Control commands to flow N2 and open Mainfolds. Cooldown reactor and purge with N2./attach Logout tagout if applicable	Tube, Quartzparts, O-Rings and T/C	Done	
		Done	
		Done	
		Done	
	Cooldown Temperature: 25°C Ramp: -2°C/min	Done	
Change the Boat (optional)	Reference Data		
Prepare polyflow for parts cleaning Remove Boat and Pedestal from loading/maintenance station. Move Boat from Reactor to Cooling Station to cool. Once Cooled remove boat though Maintenance Door.			
		Done	
		Done	
		Done	
Remove Tube	Reference Data		
Turn off cooling water Swing out reactor and disconnect water lines. Pull back spike T/C's from housing. Disconnect Paddle T/C and Remove Liner, injector and paddle T/C Remove Tube Remove all O'rings from Gas dividing ring and lower seal.		Done	
		Done	
		Done	
		Done	
		Done	
Clean Parts	Reference Data		Comments
Clean Gas dividing ring Clean Boat and Pedestal housing. Clean Liner suspension ring Clean Tube housing ring.		Done	
		Done	
		Done	
Rebuild Tube	Reference Data		Comments
Rebuild tube Rebuild Liner and components Reconnect water lines Reset spike T/Cs in housing Swing in reactor and drop onto gas dividing ring Check lift is free from set down position Install clean door plates, in the correct orientation. Use level to check, if door plate is situated correctly Insert Pedestals and boats, in correct orientation. Use level to check, if pedestal and boats are situated correctly		Done	
		Done	
		Done	
		Done	
		Done	
		Done	
		Done	
		Done	
Calibrate Boat / Test Leakrate	Reference Data		Comments
Carryout Boat Calibration for both boats (if changed) Remove Logout Tagout if applicable. Insert one boat into the reactor using the Elevator commands. Pump reactor down to standby pressure and let it stabilize for 5 mins.		Done	
Carryout Cold leak check if good continue with next step, if not, find and Resolve the issue.	Possible Problem Sources: Outgasing or Valve Error (Internal Leak), External Leak. BP: (< 10mTorr), LR: (< 5mTorr/min)		

Ramp reactor to standby temperature	SBY Temperatur: Poly(550°C), Sipos(620°C), Nitride(700°C) Ramp: 2°C/min Possible Problem Sources: Outgasing or Valve Error (Internal Leak), External Leak. BP: (< 10mTorr), LR: (< 5mTorr/min)		
Carry out hot leak check,if good continue with next step, if not, find and Resolve the issue.			
If all checks are good, batch and Queue a coating run.			
Run Coating runs on Both Boats (if changed)			
Run Pump curve			
Afterchange Tasks (Partly done By Operators)	Reference Data	Data	Comments
Run profile. Ask Operators to do this task			
Upload data to MSC	Difference Should Be the New Calibration Data on the machine		
Run Thickness test. Ask Operators to do this task			
Run Particle test. Ask Operators to do this task			
If all tests are good, change status of Si View and hand back to production.			
Issue with tool computer/MS.C.			
Comments:			